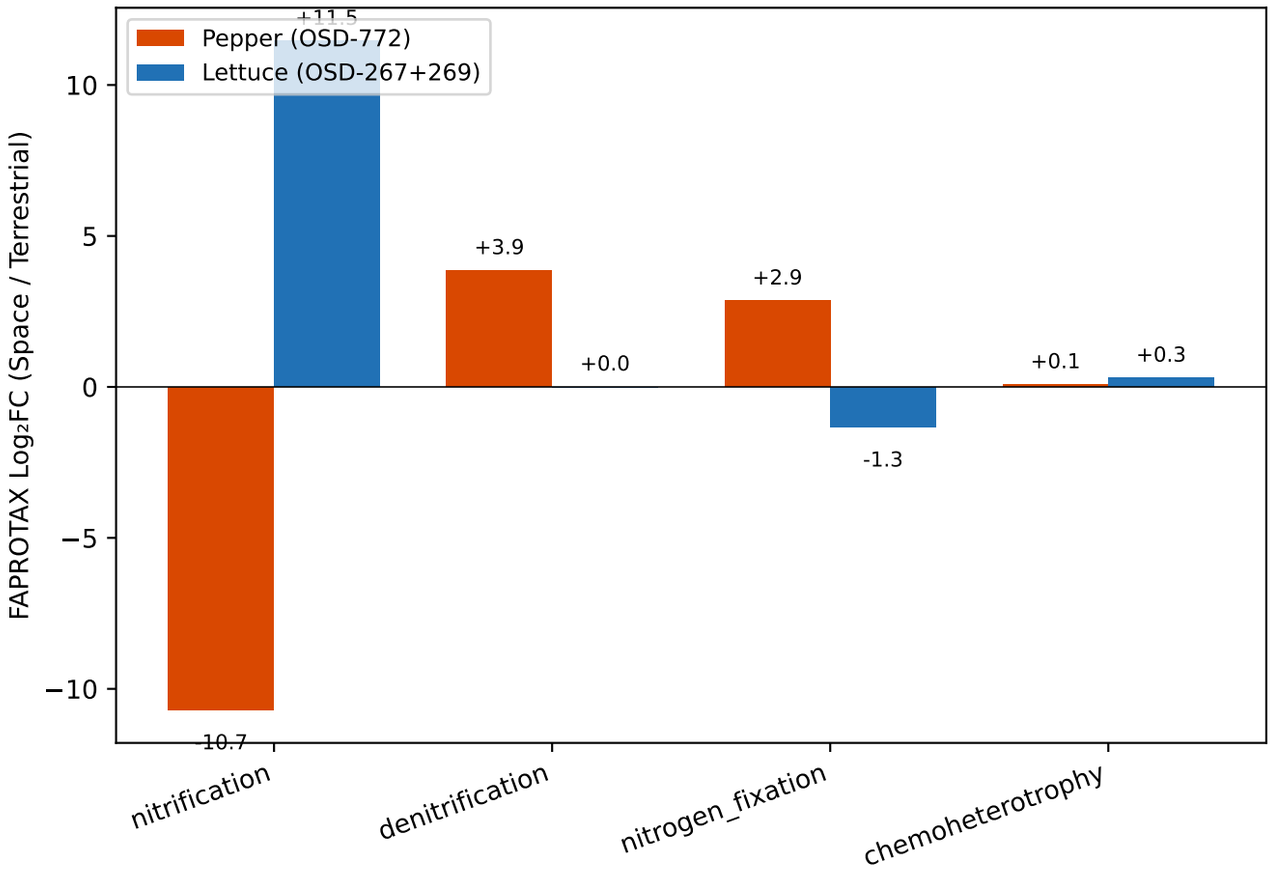
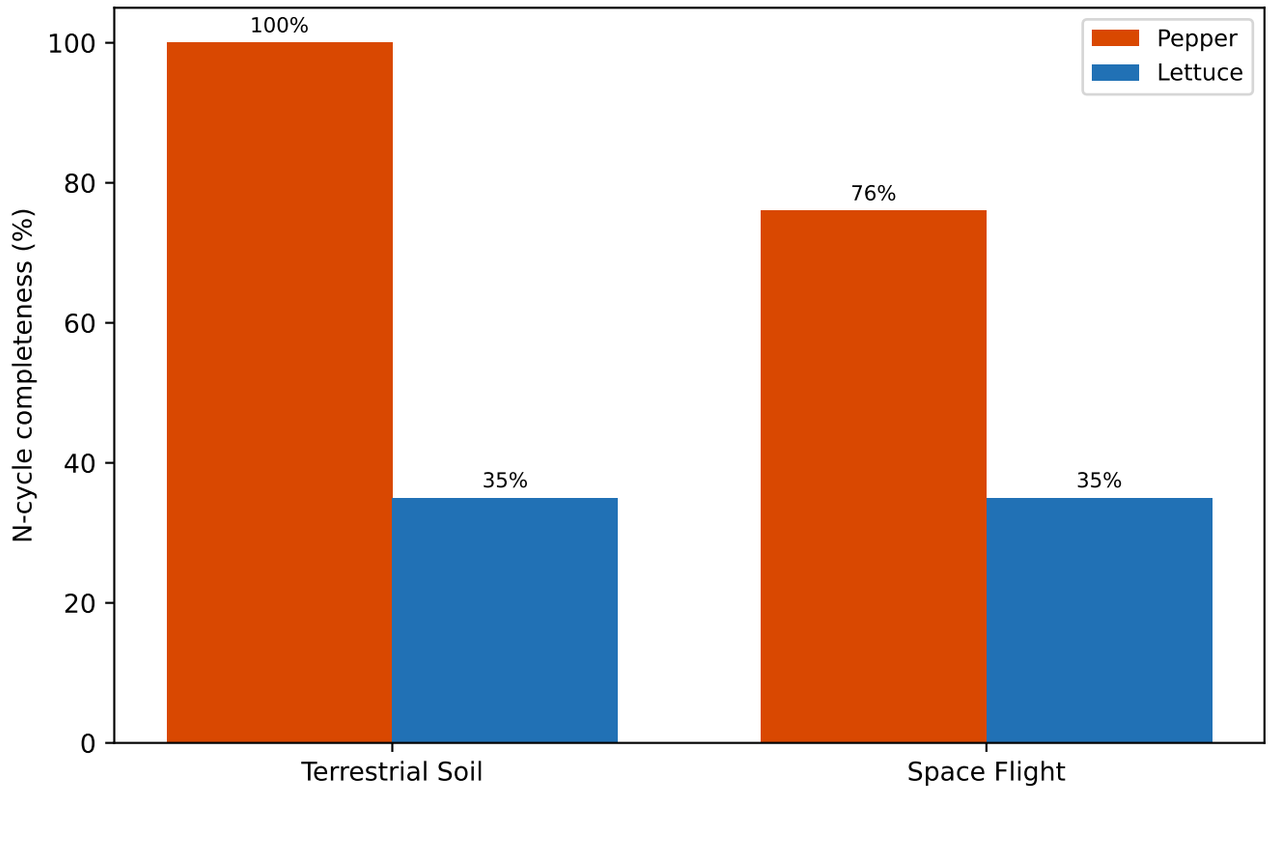
Figure 5. Crop-specific failure mode — pepper and lettuce diverge

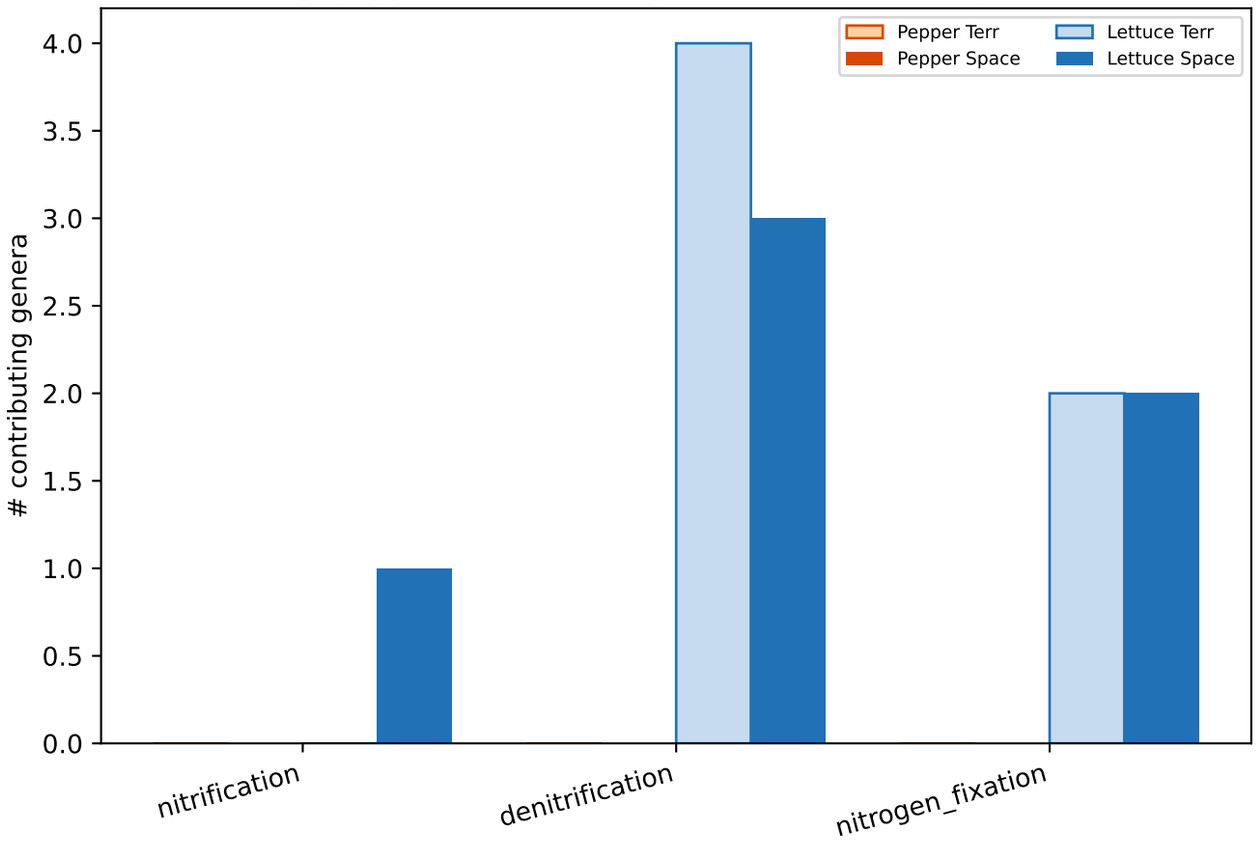
a) Divergent crop responses: pepper nitrification collapses, lettuce enriches



b) N-cycle completeness diverges between crops



c) Redundancy: pepper nitrifiers collapse (4.8→0.2), lettuce pattern differs



d) Engineering implication

Crop-specific failure modes:

- Pepper (OSD-772) shows catastrophic nitrification collapse ($Log_2FC = -10.68$, 4.8 → 0.2 genera).
- Lettuce (OSD-267+269) shows the opposite: nitrification $Log_2FC = +11.45$ in Space.
- Conclusion: BLSS biological risk models built on a single crop cannot be generalized. Per-crop microbiome profiling is required.